

An Intelligent Process Decision-Making Method for Robotic Grinding Random Defects via Incremental Learning and Database

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Abstract. Industrial robots, leveraging their high flexibility and reconfigurability, are widely applied in grinding large complex components. But for variable defects like splattered welding slags on automotive body-in-white, existing robotic grinding strategies cannot adjust dynamically, causing unstable quality. To effectively remove random defects, this paper proposes an intelligent process decision-making method via incremental learning and database. First, orthogonal experiments and F-test are employed to screen out effective modeling data. Subsequently, an incremental support vector regression (SVR) algorithm is developed to establish an upgradable robotic grinding roughness model and material removal model. On this basis, a weighted case-based reasoning (CBR) method combined with an improved multi-objective grey wolf optimizer (MOGWO) is employed to determine optimal process parameters under complex grinding conditions. Meanwhile, the residual height of weld slags after grinding is predicted by the material removal model, thereby supporting secondary grinding and reducing the need for repeated measurements. Experimental results demonstrate that random weld slags can be effectively removed. The average surface roughness (Ra) of the processed areas can reach 0.95 μm, with an average prediction error of only 9.0%, validating the effectiveness of the proposed intelligent process decision-making method.

Keywords: Robotic Grinding · Welding Slags · Materials Removal · Process Database · Intelligent Process Decision-Making

1 Introduction

Intelligent manufacturing represented by robotic machining technology has deeply integrated information technology with advanced manufacturing technologies, and is widely applied in fields such as aerospace [1], rail transit [2], and automotive manufacturing [3]. In grinding of large complex components like automotive body, robots effectively

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enhance the accuracy, efficiency, and product consistency of grinding through multi-sensor fusion, configuration of flexible end-effectors, and integration with information technology [4]. However, welding slags on automotive body feature random distribution, size variation, inconsistent grinding allowance, and variable removal difficulty. When robots are used for grinding, on the one hand, the traditional schemes with fixed paths and static process parameters are difficult to achieve efficient and high-quality grinding, and their models are not capable of being updated with the changing scenarios. On the other hand, the lack of process databases hinders systematic case management, starving robotic grinding applications for random defects of critical empirical support. Therefore, for random welding slag removal, the digital management and utilization of process cases as well as the decision-making of optimal processes are crucial to the application of robot grinding.

Extensive research is conducted in three aspects for enhancing robotic grinding quality: material removal modeling, process database development, and intelligent process decision-making.

Among model-based approaches, the material removal depth model based on the Preston equation [5–7] is the most prevalent. Hertzian contact theory [8] is also widely used in robotic grinding modeling. For example, Zhang et al. [9] established a contact stress-force model via simulation and Hertz theory, then constructed a robotic blade grinding model using multiple regression. Material removal research also involves finite element analysis [10], the tribological Archard equation [11], tool wear studies [12], and machine learning integration [13]. Surface roughness studies focus on two directions: predicting roughness from process parameters [14] and real-time sensor monitoring [15], both effective for surface quality control. While existing models fit specific grinding scenarios, they overlook material/abrasive variations, limiting application scope and reliability under changing conditions.

Fixed-scenario grinding process databases emerged earlier. For example, Choi et al. [16] and Cai et al. [17] developed GIGAS and IGA, respectively. GIGAS recommends process parameters based on models and empirical data using heuristic rules, while IGA relies on CBR for parameter recommendation. However, these databases fail to adjust parameters during machining, prompting the proposal of intelligent process databases. Wang et al. [18] developed the EconG© expert system, which integrates grinding process database and parameter recommendation modules, monitors system electrical data, and uses neural networks to optimize grinding strategies in real time. Jian et al. [19] developed an intelligent lathe monitoring system that records chatter thresholds via a database, compensates thermal expansion in real time, mitigates chatter, and enhances product quality while prolonging tool life.

Decision-making algorithms are crucial for acquiring process parameters. They can be directly used for decision-making or embedded in process databases to integrate historical case experience. Early methods relied on orthogonal experiments and statistical analysis [20], but showed limited adaptability to frequent changes. Intelligent approaches have since emerged, mainly through two perspectives: CBR [21] and process optimization [22]. While effective, these methods face limitations in addressing random welding slag defects: CBR lacks suitable process databases and struggles to adapt historical cases

to new problems; process optimization methods often fail to efficiently derive optimal solutions due to random initial parameter generation.

Thus, scenario-adaptive models, optimal-solution decision algorithms, and intelligent case-managed process databases are pivotal to overcoming the bottleneck of robotic random defect grinding.

2 Material Removal Characteristic Model for Robotic Grinding

2.1 Support Vector Regression Incremental Learning Algorithm

This study collected grinding data of four materials through orthogonal experiments: aluminum alloys 6061-T6 and 7075-T6, stainless steel 304, and carbon structural steel Q235. Subsequently, F-test is used to identify factors significantly influencing robotic grinding quality. Factors with p-values below 0.05 are regarded as significant, and their data are used to build the grinding characteristic model.

Robotic grinding is affected by numerous mutually coupled factors with complex interactions. The SVR algorithm, featuring strong generalization ability, handles complex nonlinear problems via kernel function construction. Incremental Learning (IL) updates pre-trained models with new data without full retraining, avoiding catastrophic forgetting. This paper proposes SVR-IL by integrating replay-based IL to optimize SVR:

$$\min \frac{1}{2}\|\omega\|^2 + C\sum_{i=1}^{n}\left(\zeta_i + \zeta_i^*\right) \tag{1}$$

$$s.t.\begin{cases} y_i - \omega^T\phi(x_i) - b \le \varepsilon + \zeta_i \\ \omega^T\phi(x_i) + b - y_i \le \varepsilon + \zeta_i^* \\ \zeta_i, \zeta_i^* \ge 0, i = 1, ..., n \end{cases} \tag{2}$$

where, ζ_i, ζ_i^* is the penalty term, C is the penalty coefficient, and $\phi(x_i)$ is the kernel function. The dual form is transformed as:

$$\min_{\alpha,\alpha^*} \frac{1}{2}\left(\alpha - \alpha^*\right)^T Q\left(\alpha - \alpha^*\right) + \varepsilon\left(\alpha + \alpha^*\right) - y^T\left(\alpha - \alpha^*\right) \tag{3}$$

$$s.t.\begin{cases} e^T\left(\alpha - \alpha^*\right) = 0 \\ 0 \le \alpha_i, \alpha_i^* \le C \\ i = 1, ..., n \end{cases} \tag{4}$$

where Q is a positive semidefinite matrix, and $Q_{ij} = K\left(x_i, x_j\right) = \phi(x_i)^T\phi\left(x_j\right)$. Due to the influence of abrasive belt wear in the robotic grinding process, which is time-related, the $\phi(x_i)$ is constructed as a composite kernel based on the RBF kernel and PER kernel:

$$K\left(x, x'\right) = \theta_1^2 \exp\left(-\frac{\left(x - x'\right)^2}{2\theta_2^2}\right) + \theta_3^2 \exp\left(-\frac{2}{\theta_4^2}\sin^2\left(\frac{\pi\left(x - x'\right)}{T_{\text{ker}}}\right)\right) \tag{5}$$

where, $\theta_1, \theta_2, \theta_3, \theta_4$ and T_{ker} are hyperparameters.

This paper employs sample replay, combining new samples with partial old ones for joint training during updates. The SVR-IL algorithm follows these updating steps:

(1) Compute similarity between new and old samples, then sort support vector samples and non-support vector samples in ascending order. The similarity is calculated by:

$$d_j^{(i)} = \sum_{k=1}^{m} \left| x_j^{(i)} - x_k^{new} \right| F_{kp}^T + \sum_{k=1}^{m} \left| x_j^{(i)} - x_k^{new} \right| F_h^T \tag{6}$$

where $x_j^{(i)}$ is the feature vector of the j-th sample in the i-th generation, x_k^{new} is the feature vector of the k-th sample in the new samples, F_{kp} is the feature weight vector related to k_p, and F_h is the feature weight vector related to the grinding depth. The feature weight vector consists of weight coefficients for each feature variable. The weight coefficient $F^{(i)}$ is calculated as:

$$F^{(i)} = \frac{1}{e^{p_i}} \Bigg/ \sum_{j=1}^{n} \frac{1}{e^{p_j}} \tag{7}$$

where p is calculated by the F-test between the feature variable and the target variable;

(2) Calculate the number of old samples extracted from each generation of samples:

$$M_{old}^{(n)} = \frac{M^{(n)} M_{old}}{\sum_{i=1}^{m} M^{(i)}}. \tag{8}$$

where $M_{old}^{(n)}$ is the number of samples extracted from the n-th generation, $M^{(n)}$ is the number of samples introduced when updating the n-th generation model, and M_{old} is the total number of old samples composing the training set. For each generation's extracted samples, the proportion of support vectors and non-support vectors is determined by their total number ratio.

(3) Combine new samples with top-ranked support and non-support vector samples from each generation to form the new training set X_{new}.

(4) Train the model with X_{new}, and record the support vector samples and non-support vector samples in the new samples.

2.2 Robotic Grinding Removal Depth Model and Roughness Model

The Preston equation, describing the relationship between process parameters and material removal, finds extensive applications in grinding, cutting, etc. Moreover, as the grinding wheel's outer layer is rubber and welding slag is metal, Hertzian contact condition is satisfied. Based on Preston equation, Hertzian contact theory, and SVR-IL algorithm, the robotic grinding material removal depth model is derived as:

$$H = \frac{K_p(x) F_n^a V_s^b}{V_f^c} \left[\frac{9\pi E^* (A+B)}{64 k^2 \varepsilon(k)} \right]^d \tag{9}$$

where H represents the grinding removal depth. $K_p(x)$ is the mapping model, built by the SVR - IL algorithm, relating machining feature variables to k_p (a fixed wear

coefficient).F_n is the normal grinding force, V_s is the linear velocity, V_f is the feed rate), A, B are the relative principal curvatures of the contact point, and a, b, c and d are correction coefficients. x, the input for $K_p(x)$, includes working condition features like force, velocities, grit size, belt service time, and workpiece material properties. When training $K_p(x)$, first compute k_p per data point. Then, use the F-test to check if factors significantly affecting grinding depth also impact k_p. If conditions are met, these features calculate the wear coefficient model:

$$K_p(x) = \omega_{kp}^T \phi(x) + b_{kp} \tag{10}$$

where, ω_{kp} is the weight coefficient of the variable, and b_{kp} is the bias term. By transforming (9) into a linear model and using the K_p calculated by the $K_p(x)$ to replace the original wear coefficient k_p, the linear model of the removal depth model is constructed by combining with other inputs as training samples:

$$\ln H = a \ln F_n + b \ln V_s - c \ln V_f + d \ln \left[\frac{9\pi E^*(A+B)}{64k^2 \varepsilon(k)} \right] + \ln K_p \tag{11}$$

During training, the loss function is defined based on the incremental learning regularization paradigm as:

$$\begin{aligned} L = \alpha \frac{1}{n} \sum_{i=0}^{n} \left(\omega_{new}^T x_{new} - y_{new} \right)^2 + \beta \frac{1}{n} \sum_{i=0}^{n} \left(\omega_{old}^T x_{new} - y_{new} \right)^2 \\ + \lambda \left(\| \omega_{new} \|^2 + \| \omega_{old} \|^2 + \| \omega_{new} - \omega_{old} \|^2 \right) \end{aligned} \tag{12}$$

where ω_{new} is the coefficient matrix of the new model, ω_{old} is the coefficient matrix of the old model, x_{new} is the feature vector of the training samples, y_{new} is the actual output of the training samples, λ is the regularization coefficient, and α, β are the incremental coefficient, with the calculation formula:

$$\begin{cases} \alpha = \dfrac{M_{old}}{M_{old} + M_{new}} \\ \beta = \dfrac{M_{new}}{M_{old} + M_{new}} \end{cases} \tag{13}$$

where M_{old} is the total number of old samples and M_{new} is the total number of new samples. The coefficients a, b, c, and d are obtained via stochastic gradient descent training, and substituting them into (9) yields the modified removal depth model.

Due to the lack of a universal physical model for surface roughness, this paper directly employs the SVR-IL algorithm to fit the robotic grinding surface roughness model. The modeling steps are shown in Fig. 1.

Using the F-test, factors significantly influencing surface roughness are identified and integrated into input features. SVR hyperparameters (C, ε, Γ) and kernel function parameters ($\theta_1, \theta_2, \theta_3, \theta_4, T_{ker}$) are optimized via random search with predefined ranges. Samples are split 80%/20% for training/testing with 50 search iterations. All samples

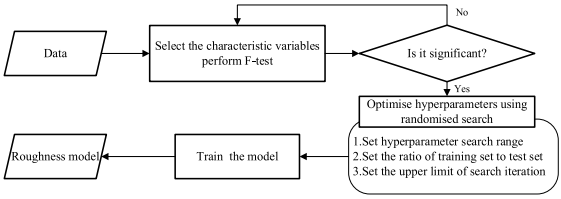


Fig. 1. Robotic grinding surface roughness modelling process.

are input into the training process to obtain the surface roughness model, which can be expressed as:

$$R_a(x) = \omega_{R_a}^T \phi_{R_a}(x) + b_{R_a} \tag{14}$$

where, ω_{R_a} represents the feature weights of the model, $\phi_{R_a}(x)$ denotes the kernel function, and b_{R_a} is the bias term. When updating the model, the optimal hyperparameter combination is also searched according to the original settings before training.

3 Intelligent Process Decision-Making Based on Database

After deriving the robotic grinding model, intelligent optimizers recommend process parameters, but their stochastic nature challenges parameter regulation. This paper establishes a case database and proposes a hybrid method integrating weighted-CBR and MOGWO. Comprising four steps—case retrieval, reuse, revision, and learning—it uses Analytic Hierarchy Process (AHP) for factor weights, weighted CBR for case matching, improved MOGWO for optimization, and updates the model by adding new cases.

The weighted CBR method constructs a hierarchical feature model, builds a judgment matrix using F-test p-values to calculate feature weights, normalizes them via geometric mean, and validates consistency (CR < 0.1) to achieve weighted case matching and optimization. When historical cases retrieved via case matching fail to meet machining requirements for new process problems, case revision is performed. This paper enhances the GWO for robotic grinding parameter optimization, with steps as follows:

(1) Initial population: To enhance efficiency, optimized process parameters are extracted from the top 10 case-based reasoning cases to form individual positions. Remaining population positions are determined by:

$$x = \sum_{i=1}^{m} \alpha_i x_i + \frac{1}{2} x_{rand} \tag{15}$$

where,x_i is the individuals derived from case-based reasoning,α_i denotes random numbers within [0, 1) that sum to 1, and x_{rand} is a vector randomly generated within the value range.

(2) Fitness function: To ensure the surface quality of the workpiece after grinding, the removal depth and surface roughness are taken as the optimization objectives, so the

fitness function is modified as:

$$\begin{aligned} \min fitness_h &= h_{\exp} - H(X) \\ \min fitness_{Ra} &= R_a(X) - Ra_{\exp} \end{aligned} \tag{16}$$

where, $h_{\exp}$ is the target removal depth, $Ra_{\exp}$ is the expected surface roughness, $H(X)$ and $R_a(X)$ are the established robotic grinding removal depth model and surface roughness model, respectively. X is the input vector of the model formed by combining the population individual x and the existing descriptions of the new process problem.

(3) Multi-objective Ranking: Non-dominated sorting of NSGA-II[23] efficiently solves each layer of the Pareto front. However, direct integration of NSGA-II into GWO has three drawbacks: crowding distances set to infinity for extreme values lead to local optima; randomness occurs in α wolf selection when there are only two solutions in the Pareto front; priorities of optimization objectives are not considered in practical applications. To address the above issues, this study optimizes the multi-objective ranking method: 1) Normalize fitness values of each individual; 2) Use fast non-dominated sorting on normalized values to derive non-dominated frontiers; 3) Compute comprehensive SCORE for each individual; 4) Sort frontiers by ascending SCORE. The comprehensive *SCORE* is calculated as follows:

$$SCORE_i = e^{1-t} f_h^i + e^{t-1} f_{Ra}^i \tag{17}$$

where,f_h is the normalized value of $fitness_h$,f_{Ra} is the normalized value of $fitness_{Ra}$, and t is the machining factor. Set t = 0 for rough machining (focused on allowance removal) and t = 1 for finish machining (demanding lower surface roughness). The formula balances sample diversity and machining requirements, offering an optimal ranking criterion for robotic grinding optimization.

4 Experimental Verification

4.1 Experimental Setup

To validate the proposed intelligent decision-making method for robotic grinding, a random welding slag robotic grinding experimental platform was constructed as shown in Fig. 2. System hardware includes a scanner (PowerScan-Pro2.3M), roughness tester (Mitutoyo SJ-210), force sensor (ATI Omega160 SI-1500-240), and industrial robot (ABB IRB 6700-200/2.6). Software components comprise RobotStudio simulation software, PowerPac path planning plug-in, TestSignalViewer force monitoring software, and the developed process database.

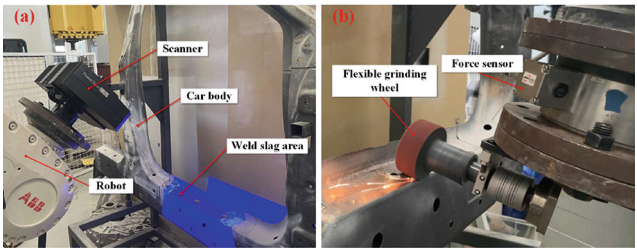


Fig. 2. Random defects intelligent process decision grinding experimental platform. (a) Weld slag measurement, (b) grinding experiments.

4.2 Preparation for Grinding

To verify the grinding effect of robot on welding slags of varying heights, the slag point cloud is secondarily clustered using the method in [24] based on position and height. Machining areas are divided, and the shortest path is solved via an improved genetic algorithm to derive the grinding trajectory in Fig. 3, where regions denote to-be-ground areas classified by maximum slag height.

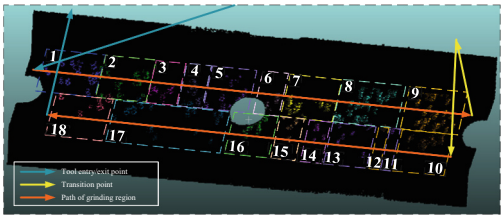


Fig. 3. Weld slag clustering and trajectory planning result.

As shown in Fig. 4, the measured maximum slag height and other factors are fed into the process database, which performs concurrent case and model matching. Matched models predict grinding quality: parameters are applied directly if predictions meet requirements; otherwise, case revision optimizes them. The optimized strategy updates the case library.

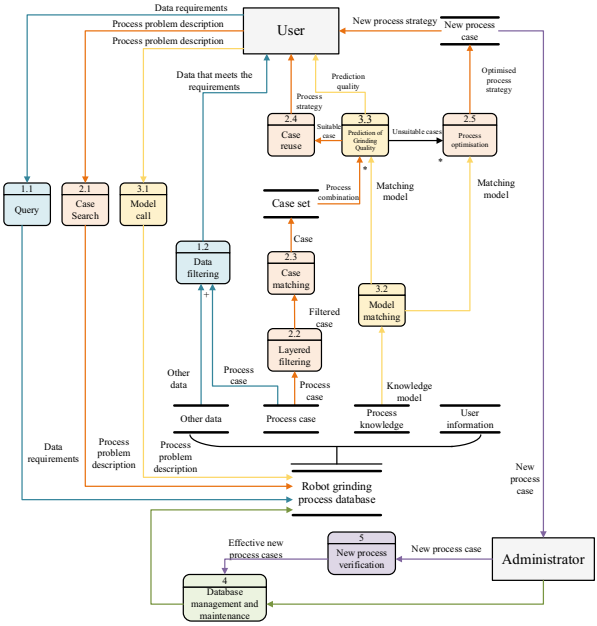


Fig. 4. Process parameter decision-making data flow.

The decision results specify two grinding passes: the first uses a 120-grit abrasive belt for roughing, with parameters and predicted results detailed in Table 1. The second machining is determined by rough machining predictions. As shown, machining areas 1, 3, 4 have incomplete slag removal, and all areas have excessive roughness. A 240-grit abrasive belt is used for fine machining, with parameters in Table 2.

Table 1. Rough grinding process decision-making results.

Region	Maximum slag height (mm)	Process parameters			H_{pre}(mm)	Ra_{pre}(μm)
		F_N(N)	V_s(m/s)	V_f(mm/s)		
1	0.561	20.1	8.8	23.3	0.489	1.646
2	0.399	11.5	8.5	7.9	0.549	1.998
3	0.573	26.3	15.1	12.9	0.507	2.132
4	0.673	29.2	12.9	18.3	0.645	1.941
5	0.426	20.0	7.6	6.8	0.638	2.040
6	0.673	15.1	15.7	18.6	0.703	1.936
7	0.556	33.3	10.1	9.7	0.629	2.017
8	0.624	18.4	8.5	12.8	0.661	1.967
9	0.538	11.2	8.5	8.0	0.756	1.999
10	0.673	10.8	11.3	13.0	0.681	1.990

(*continued*)

Table 1. (*continued*)

Region	Maximum slag height (mm)	Process parameters			H_{pre}(mm)	Ra_{pre}(μm)
		F_N(N)	V_s(m/s)	V_f(mm/s)		
11	0.292	9.3	15.6	19.0	0.371	1.877
12	0.673	33.5	12.3	12.5	0.726	2.058
13	0.292	8.9	20.3	16.0	0.293	1.997
14	0.340	29.7	6.1	23.5	0.341	1.636
15	0.147	14.0	9.7	11.1	0.261	2.022
16	0.515	24.2	11.8	21.4	0.558	1.817
17	0.469	19.5	10.8	14.5	0.521	1.966
18	0.422	19.8	9.9	9.8	0.461	2.080

Table 2. Fine grinding process decision-making results.

Region	Weld slag allowance (mm)	Process parameters			Ra_{pre}(μm)
		F_N(N)	V_s(m/s)	V_f(mm/s)	
1	0.072	16.6	25.2	22.8	0.960
3	0.066	23.5	17.6	26.6	0.935
4	0.028	13.3	17.6	32.3	0.909
Else	0	14.8	13.2	31.3	0.844

4.3 Analysis of Experimental Results

As depicted in Fig. 5(a), the pre-machining automotive body is covered with numerous welding slags, with the slag heights being notably higher in the head and tail regions. Figure 5(b) illustrates the post-grinding condition, where randomly distributed small slags have been entirely removed; however, partial residues persist in machining areas 1, 10, and 17. The post-grinding slag point cloud is presented in Fig. 5(c). In machining area 1, the remaining slag allowance is relatively substantial, reaching a maximum height of 0.1315 mm, whereas the heights of other residual slags are all below 0.1 mm.

In machining areas 1 and 10, significant force fluctuations at initial contact cause motor power to exceed the rated value, reducing rotation speed below the set value and weakening grinding ability, leading to slag residues. In area 17, an uneven body bottom and calibration errors prevent the grinding wheel from clinging to the workpiece, causing local insufficient grinding. However, these partial residues do not affect the verification of grinding process decision results.

In all machining areas obtained by slag point cloud clustering, the surface roughness was measured. During measurement, a 0.8 mm line segment is randomly sampled

along the tangential direction of the grinding wheel linear velocity. In areas 1, 3, and 4 (incompletely slag-removed), three random positions were measured and averaged. For fully removed areas, three points at the front, middle, and end of the path were measured and averaged. Results are shown in Table 3.

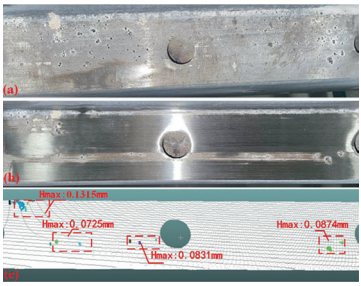


Fig. 5. Comparison of welding slag region before and after grinding. (a) Welding slag area before grinding, (b) welding slag area after grinding, (c) point cloud of welding slag area after grinding.

Table 3. Surface roughness measurement results of grinding surface.

Grinding area	Predicted Ra (μm)	Measured Ra (μm)	Errors
1	0.960	1.1	13.1%
3	0.935	1.0	4.2%
4	0.909	0.8	7.5%
Incomplete removal areas	0.844	1.0	17.1%
Other areas	0.844	0.8	2.9%

The experimental measurements generally align with the predicted results. For the normally removed slag, the maximum error is 7.5% and the minimum is 2.9%; for the incompletely removed parts due to equipment limitations, the maximum error is 17.1% and the minimum is 13.1%, with an average roughness error of 9.0%.

When comparing the actual slag removal height and surface roughness with the predicted values, although experimental errors result in incomplete slag removal, the results validate that the established material removal characteristic model for robotic grinding of vehicle body welding slag can guide grinding processes. This demonstrates that the proposed intelligent process decision-making method for robotic grinding is applicable to practical robotic grinding operations.

5 Conclusion

This study presents an intelligent robotic grinding decision method via incremental learning and self-updating database, enabling adaptation to random defects, optimal parameter delivery, and continuous updating. Experimental verification is carried out

for the robotic grinding and removal scenario of automotive body welding slags, and the following conclusions are drawn:

(1) The SVR-IL-based material removal and roughness models can not only reflect the relationship between multiple factors and machining quality, but also enable continuous learning, addressing the issue of significant application scenario limitations in traditional models.
(2) The database-driven intelligent decision-making method integrates weighted CBR and improved MOGWO for case matching and parameter optimization, resolving limitations of traditional methods—their inflexibility and suboptimal machining performance under complex conditions.
(3) The robotic grinding experiment on automotive welding slags validates the method: random defects are effectively removed, post-machining roughness (Ra) stays within 1.2 μm (avg. 0.95 μm), and the 9.0% average prediction error confirms model accuracy and decision-making feasibility.

The method applies to automotive body local defects repair and can be adapted for other scenarios via simple modifications, such as 3D putty or paint film defects on high-speed railway white body. However, the accuracy of tool calibration requires improvement. Future work will focus on enhancing tool calibration to improve machining stability.

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